Name: Gaddale Charmi Bai

Roll Number: 23071A0516

AIM: Implementation of Linear Regression Model

Dataset: housing-dataset

```
#Importing important Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#downloading Dataset
import kagglehub
path = kagglehub.dataset_download("ashydv/housing-dataset")
```

```
Warning: Looks like you're using an outdated `kagglehub` version (installed: 0.3.13), please consider upgrading to the lates
Downloading from https://www.kaggle.com/api/v1/datasets/download/ashydv/housing-dataset?dataset_version_number=1...
100%|██████████| 4.63k/4.63k [00:00<00:00, 13.4MB/s]Extracting files...
```

```
#printing the path
print(path)
```

```
/root/.cache/kagglehub/datasets/ashydv/housing-dataset/versions/1
```

```
import os
os.listdir(path)
```

```
['Housing.csv']
```

```
#Loading and Reading DataSet
df=pd.read_csv(path+"/Housing.csv")
```

```
print(df.head())
print(df.info())
```

```
      price  area  bedrooms  bathrooms  stories  parking  HighPrice  \
0  13300000  7420         4          2        3        2          1
1  12250000  8960         4          4        4        3          1
2  12250000  9960         3          2        2        2          1
3  12215000  7500         4          2        2        3          1
4  11410000  7420         4          1        2        2          1

   mainroad_yes  guestroom_yes  basement_yes  hotwaterheating_yes  \
0          True          False         False                False
1          True          False         False                False
2          True          False          True                False
3          True          False          True                False
4          True           True          True                False

   airconditioning_yes  prefarea_yes  furnishingstatus_semi-furnished  \
0                 True          True                            False
1                 True         False                            False
2                False          True                             True
3                 True          True                            False
4                 True         False                            False

   furnishingstatus_unfurnished
0                         False
1                         False
2                         False
3                         False
4                         False
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 15 columns):
 #   Column                           Non-Null Count  Dtype
---  ------                           --------------  -----
 0   price                            545 non-null    int64
```

```
 1   area                             545 non-null    int64
 2   bedrooms                         545 non-null    int64
 3   bathrooms                        545 non-null    int64
 4   stories                          545 non-null    int64
 5   parking                          545 non-null    int64
 6   HighPrice                        545 non-null    int64
 7   mainroad_yes                     545 non-null    bool
 8   guestroom_yes                    545 non-null    bool
 9   basement_yes                     545 non-null    bool
 10  hotwaterheating_yes              545 non-null    bool
 11  airconditioning_yes              545 non-null    bool
 12  prefarea_yes                     545 non-null    bool
 13  furnishingstatus_semi-furnished  545 non-null    bool
 14  furnishingstatus_unfurnished     545 non-null    bool
dtypes: bool(8), int64(7)
memory usage: 34.2 KB
None
```

```
#Feature Selection
X=df[["area","bedrooms","bathrooms","stories","parking"]]
y=df["price"]
```

```
#Training and Testing
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

```
#Model Selection
model=LinearRegression()
```

```
#Model Training
model.fit(X_train,y_train)
```

```
▾ LinearRegression i ?
LinearRegression()
```

```
#Prediction
y_pred=model.predict(X_test)
```

```
print("Model Coefficients:",model.coef_)
print("Model Intercept:",model.intercept_)
```

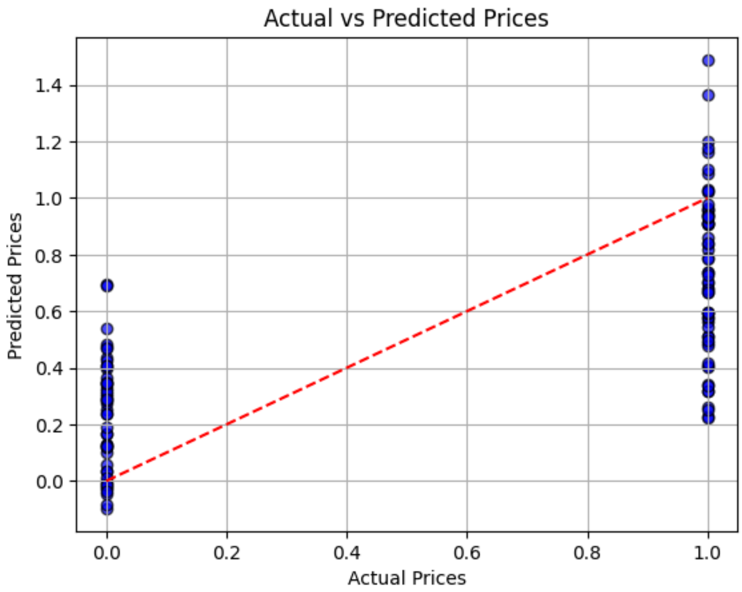
```
Model Coefficients: [ 6.76726480e-05  4.20660584e-02  1.35420828e-01  9.90543616e-02
  1.81600596e-02  1.41727904e-01  1.48572786e-01  1.47661408e-01
  9.24802498e-02  1.92010345e-01  1.33630460e-01  7.37594938e-02
 -8.86396194e-02]
Model Intercept: -0.6404462536034508
```

```
#Evaluation
mse=mean_squared_error(y_test,y_pred)
rmse=mse**0.5
mae=mean_absolute_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(f"MSE:{mse:.2f}")
print(f"RMSE:{rmse:.2f}")
print(f"MAE:{mae:.2f}")
print(f"R2 Score:{r2:.2f}")
```

```
MSE:0.13
RMSE:0.35
MAE:0.29
R2 Score:0.49
```

```
#Visuallization
plt.scatter(y_test, y_pred, color="blue", edgecolor="k", alpha=0.7)

plt.plot(
    [min(y_test), max(y_test)],
    [min(y_test), max(y_test)],
    color="red",
    linestyle="--"
```

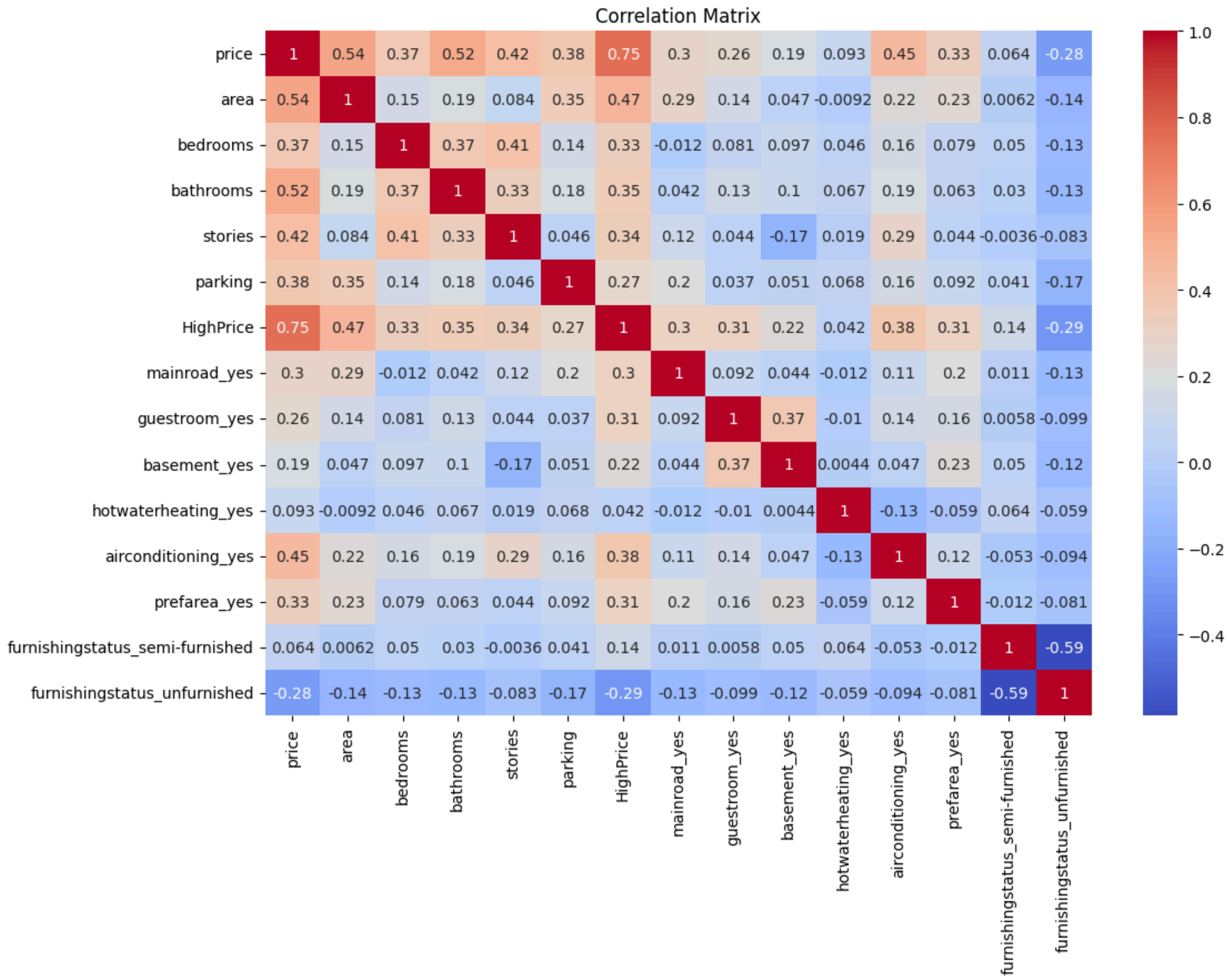
```
)

plt.title("Actual vs Predicted Prices")
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.grid(True)
plt.show()
```




```
corr_matrix = df.corr()
print(corr_matrix)
```

```
area                                    -0.142278
bedrooms                                -0.126252
bathrooms                               -0.132107
stories                                 -0.082972
parking                                 -0.165705
HighPrice                               -0.288103
mainroad_yes                            -0.133123
guestroom_yes                           -0.099023
basement_yes                            -0.117935
hotwaterheating_yes                     -0.059194
airconditioning_yes                     -0.094086
prefarea_yes                            -0.081271
furnishingstatus_semi-furnished         -0.588405
furnishingstatus_unfurnished             1.000000
```

```
plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```
print(df.corr()["price"].sort_values(ascending=False))
```

```
price                  1.000000
HighPrice              0.747069
area                   0.535997
bathrooms              0.517545
airconditioning_yes    0.452954
stories                0.420712
parking                0.384394
bedrooms               0.366494
prefarea_yes           0.329777
mainroad_yes           0.296898
```

```
guestroom_yes                      0.255517
basement_yes                       0.187057
hotwaterheating_yes                0.093073
furnishingstatus_semi-furnished    0.063656
furnishingstatus_unfurnished      -0.280587
Name: price, dtype: float64
```